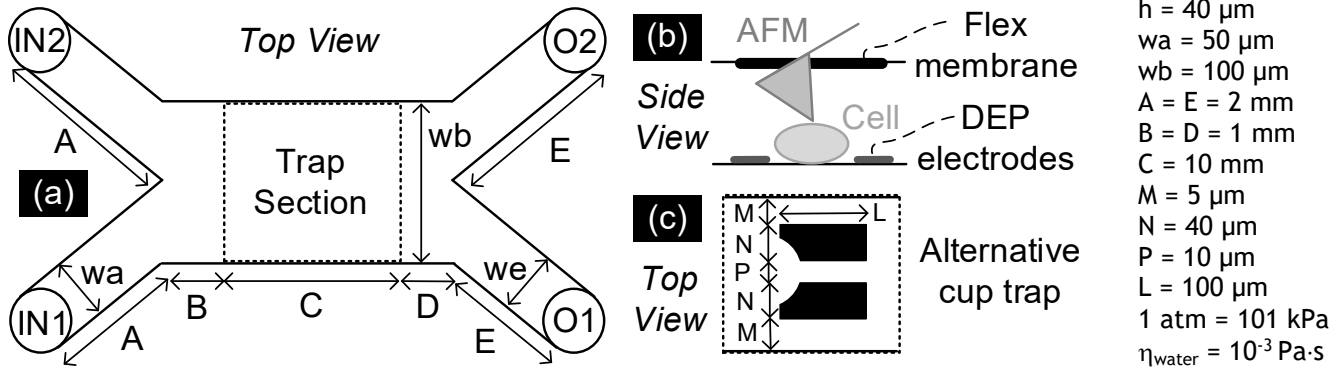


Politecnico di Milano
Biochip A.Y. 2022/2023 - M. Carminati
 Exam of 1.09.2023

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

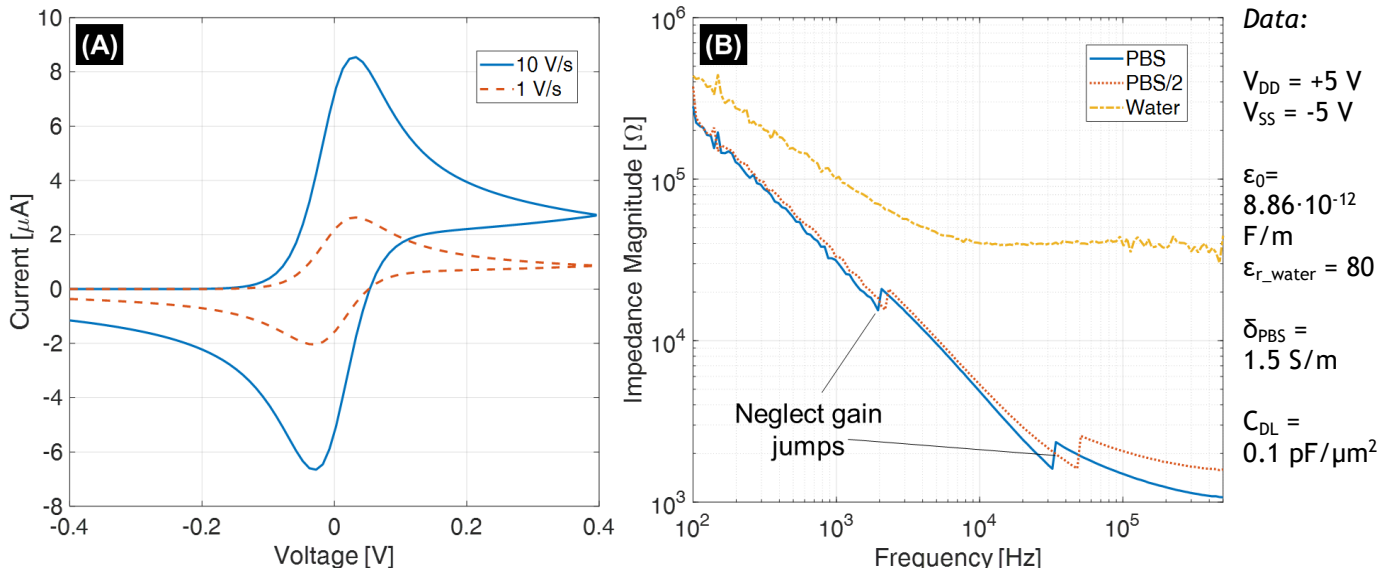
Consider the following microfluidic device to trap a single cell and run AFM imaging of the cell surface in liquid. Movement of the AFM tip on top of the cell is allowed by a flexible membrane on the ceiling of the trap section. Two types of traps can be used: a dielectrophoretic trap (b) or a cup trap (c). The height is constant equal to h .



- Draw the complete electrical fluidic equivalent of the system with the DEP trap (b) assuming all filled with water and neglecting the presence of the AFM tip.
- Assuming $P_{IN1} = 3 \text{ atm}$ and $P_{IN2} = 2 \text{ atm}$ and open outlets, find the speed of water in all sections. If inlets and outlets are swapped, how does the solution change?
- If $Q_{IN1} = 2 \mu\text{L/min}$ and $Q_{IN2} = 1 \mu\text{L/min}$, draw the separation of flows in the rest of the sections.
- Draw a possible configuration of planar electrodes to realize the DEP trap (b). Is it based on pDEP or nDEP?
- Describe the working principle of the Atomic Force Microscope (draw a scheme).
- Now consider variant (c): find the new equivalent circuit (approx). Compare the pros and cons of the two trap types.

Exercise 2

Now consider the following electrochemical measurements performed with a planar **gold** working electrode and neglect the other (counter) electrode:



- What technique is (A) and what information can be extracted from plot A? What are the numbers in the legend?
- What information can be extracted from plot (B)? Draw the corresponding impedance model for the case of PBS. What is the conductivity of water (dashed spectrum) here?
- Size a **current amplifier** able to perform both (A) and (B) considering current ranges and bandwidths.
- Study the noise of the proposed circuit.
- If **impedance spectroscopy** is run at a speed of 1 ms per freq. point, what should be the lock-in bandwidth?
- Redraw the PBS spectrum of plot (B) in a **Cole Cole plot**.

Multichance students with 30% reduction of the exercises please skip points e) and f) of both exercises.